

## ST CROIX LIME TREE BAY, Virgin Islands, U.S.

WMO: 997349

Lat: 17.695N

Lon: 64.754W

Elev: 33

StdP: 14.68

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 73.9       | 74.8 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 18.4                  | 78.6 | 16.8 | 78.9 | 7.4                   | 80   | 0.411 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 5.8                    | 88.2            | N/A  | 87.7 | N/A  | 87.1 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 7.8                  | 130  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 17.7          | 15.7 | 14.2 | DB                         | 71.5 | 89.2               | 0.8 | 0.8                                                | 70.9 | 89.8       | 70.4 | 90.2       | 70.0 | 90.6       | 69.4 | 91.2 | (4) |
| (5)               |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|----------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                            | 81.5   | 78.8  | 78.5  | 78.8  | 80.0  | 81.5  | 83.2  | 83.5  | 84.0  | 83.9  | 83.5  | 81.8  | 80.0  |
|                                                                                  | DBStd                            | 2.47   | 1.14  | 1.05  | 1.57  | 1.38  | 1.63  | 1.21  | 1.15  | 1.45  | 1.31  | 1.57  | 1.50  | 1.30  |
|                                                                                  | HDD50                            | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD65                            | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD50                            | 11488  | 893   | 799   | 892   | 901   | 977   | 995   | 1038  | 1055  | 1016  | 1038  | 954   | 930   |
|                                                                                  | CDD65                            | 6013   | 428   | 379   | 427   | 451   | 512   | 545   | 573   | 590   | 566   | 573   | 504   | 465   |
|                                                                                  | CDH74                            | 65260  | 3483  | 2989  | 3487  | 4302  | 5618  | 6599  | 7078  | 7479  | 7153  | 7123  | 5574  | 4375  |
| CDH80                                                                            | 18486                            | 264    | 173   | 311   | 619   | 1390  | 2316  | 2647  | 3039  | 2863  | 2723  | 1496  | 645   |       |
| Wind                                                                             | WSAvg                            | 7.8    | 7.8   | 7.8   | 7.7   | 8.1   | 7.9   | 8.5   | 8.9   | 8.4   | 6.8   | 7.2   | 7.0   | 7.7   |
| Precipitation                                                                    | PrecAvg                          | 40.0   | 2.2   | 1.6   | 1.5   | 2.6   | 4.1   | 2.3   | 2.8   | 3.8   | 5.0   | 4.8   | 5.7   | 4.2   |
|                                                                                  | PrecMax                          | 81.3   | 7.5   | 3.3   | 5.3   | 6.2   | 17.0  | 7.4   | 8.3   | 12.6  | 18.7  | 15.7  | 20.1  | 11.6  |
|                                                                                  | PrecMin                          | 21.4   | 0.0   | 0.2   | 0.0   | 0.4   | 0.4   | 0.2   | 0.8   | 0.8   | 1.3   | 0.3   | 1.5   | 0.4   |
|                                                                                  | PrecStd                          | 12.8   | 1.7   | 1.0   | 1.2   | 1.6   | 4.5   | 2.1   | 1.7   | 3.0   | 3.7   | 3.3   | 4.8   | 2.8   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                             | DB     | 83.4  | 82.9  | 84.6  | 85.5  | 87.1  | 87.7  | 88.1  | 90.2  | 88.9  | 88.8  | 86.9  | 85.4  |
|                                                                                  |                                  | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 2%                               | DB     | 82.6  | 82.3  | 83.3  | 84.3  | 85.8  | 87.0  | 87.3  | 88.5  | 88.0  | 87.9  | 86.2  | 84.3  |
|                                                                                  |                                  | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5%                               | DB     | 82.0  | 81.7  | 82.3  | 83.5  | 85.1  | 86.3  | 86.7  | 87.5  | 87.3  | 87.2  | 85.5  | 83.5  |
|                                                                                  |                                  | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 10%                              | DB     | 81.5  | 81.1  | 81.7  | 82.7  | 84.4  | 85.8  | 86.2  | 86.8  | 86.7  | 86.5  | 85.0  | 82.8  |
|                                                                                  |                                  | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 0.4%                             | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |                                  | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 2%                               | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |                                  | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB                            | MDBR   | 5.7   | 5.5   | 5.6   | 5.4   | 5.3   | 5.1   | 5.6   | 5.8   | 5.9   | 5.9   | 6.2   | 6.0   |
|                                                                                  |                                  | MCDBR  | 5.9   | 5.9   | 5.8   | 5.7   | 5.5   | 5.3   | 6.0   | 6.0   | 6.2   | 6.2   | 6.3   |       |
|                                                                                  |                                  | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |       |
|                                                                                  | 5% WB                            | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |                                  | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | Clear-Sky<br>Solar<br>Irradiance | taub   | 0.370 | 0.375 | 0.380 | 0.405 | 0.449 | 0.514 | 0.518 | 0.478 | 0.437 | 0.392 | 0.378 | 0.374 |
| taud                                                                             |                                  | 2.583  | 2.563 | 2.545 | 2.496 | 2.386 | 2.163 | 2.147 | 2.299 | 2.440 | 2.587 | 2.620 | 2.607 |       |
| Ebn at Noon                                                                      |                                  | 291    | 295   | 295   | 286   | 270   | 251   | 250   | 262   | 274   | 286   | 287   | 286   |       |
| Edh at Noon                                                                      |                                  | 30     | 32    | 33    | 35    | 39    | 48    | 49    | 43    | 36    | 31    | 29    | 29    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg                           | 1569   | 1825  | 2011  | 2106  | 2032  | 2043  | 2078  | 2050  | 1914  | 1694  | 1500  | 1453  |       |
|                                                                                  | RadStd                           | 37     | 33    | 70    | 108   | 124   | 115   | 92    | 91    | 116   | 114   | 95    | 65    |       |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (7 neighbors) | +0.41 | +0.81   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +142  | +144  |

Nomenclature: See separate page